

IGCSE Mathematics 0580 — Formula Sheet (Extended)

Cambridge IGCSE 0580, exams 2025–2027. Mensuration formulae are printed in the exam; everything beyond that must be memorised.

NUMBER

Percentage change	$= (\text{new} - \text{original}) / \text{original} \times 100$
Reverse percentage	$\text{original} = \text{new} \div (1 + \text{rate}) \quad \text{e.g. after } +20\%: \text{divide by } 1.2$
Compound interest	$\text{final} = P(1 + r/100)^n$
Simple interest	$I = P \cdot r \cdot t / 100$
Speed, density, pressure	$v = d/t; \quad \rho = m/V; \quad p = F/A$
Index laws	$a^p \times a^q = a^{p+q}; \quad a^p \div a^q = a^{p-q}; \quad (a^p)^q = a^{pq}$
Negative & fractional	$a^{-p} = 1/a^p; \quad a^{1/2} = \sqrt{a}; \quad a^{p/q} = (\sqrt[q]{a})^p$
Standard form	$A \times 10^n, \text{ where } 1 \leq A < 10$

ALGEBRA

Quadratic formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Difference of squares	$a^2 - b^2 = (a - b)(a + b)$
nth term (linear)	$T_n = an + b$
nth term (quadratic)	$T_n = an^2 + bn + c$
Composite functions	$fg(x) = f(g(x)) \text{ — apply } g \text{ FIRST}$
Inverse function	$f^{-1}: \text{swap } x \text{ \& } y \text{ in } y = f(x), \text{ make } y \text{ the subject}$

MENSURATION (given in exam — know when to use)

Area	rectangle: lw ; triangle: $\frac{1}{2}bh$; parallelogram: bh ; trapezium: $\frac{1}{2}(a+b)h$
Circle	$A = \pi r^2; \quad C = 2\pi r = \pi d$
Sector	$\text{arc} = \left(\frac{\theta}{360}\right) \times 2\pi r; \quad \text{area} = \left(\frac{\theta}{360}\right) \times \pi r^2$
Prism volume	$V = \text{area of cross-section} \times \text{length}$
Cylinder	$V = \pi r^2 h; \quad \text{curved SA} = 2\pi rh$
Cone	$V = \frac{1}{3}\pi r^2 h; \quad \text{curved SA} = \pi rl \quad (l = \text{slant height})$
Sphere	$V = \frac{4}{3}\pi r^3; \quad SA = 4\pi r^2$

GEOMETRY & CIRCLE THEOREMS — learn the reasons word-for-word

Polygon angles	$\text{interior sum} = (n - 2) \times 180^\circ; \quad \text{exterior} = 360^\circ/n$
Pythagoras	$a^2 + b^2 = c^2 \text{ (right angle only)}$
Similar shapes	$\text{lengths } \propto k; \quad \text{areas } \propto k^2; \quad \text{volumes } \propto k^3$
Centre & circumference	$\text{angle at centre} = 2 \times \text{angle at circumference}$
Same segment	$\text{angles in the same segment are equal}$
Semicircle	$\text{angle in a semicircle} = 90^\circ$
Cyclic quadrilateral	$\text{opposite angles sum to } 180^\circ$
Tangent	$\text{tangent } \perp \text{ radius}$
Alternate segment	$\text{angle between tangent \& chord} = \text{angle in alternate segment}$

TRIGONOMETRY

SOHCAHTOA	$\sin = \text{opp/hyp}; \quad \cos = \text{adj/hyp}; \quad \tan = \text{opp/adj}$
Sine rule	$a/\sin A = b/\sin B = c/\sin C$
Cosine rule	$c^2 = a^2 + b^2 - 2ab \cdot \cos C$
Triangle area	$A = \frac{1}{2}ab \cdot \sin C$

Exact values

$$\sin 30 = \frac{1}{2}, \cos 60 = \frac{1}{2}, \tan 45 = 1, \sin 60 = \frac{\sqrt{3}}{2}, \tan 30 = \frac{1}{\sqrt{3}}$$

STATISTICS & PROBABILITY

Mean

$$\text{list: } \frac{\sum x}{n}; \quad \text{table: } \frac{\sum fx}{\sum f}$$

Probability

P = favourable/total; all outcomes sum to 1

Independent events

$$P(A \text{ and } B) = P(A) \times P(B)$$

Conditional (tree)

$$P(B|A) = P(A \text{ and } B)/P(A) \text{ — denominator changes!}$$